

Andrew Scutt

EDUCATIONAL BACKGROUND

University of Toronto

Toronto, Ontario
Sept. 2022 – Jun 2027

- Psychology Research Specialist
- Economics Major with a Focus in Data Analytics
- Mathematics Minor

HONORS AND AWARDS

Sixth Annual Tech Econference Final Round Paper	April 2026
Economics Undergraduate Research Grant (830\$)	December 2025
Fifth Annual Tech Econference Final Round Paper	April 2025
Undergraduate Research Fund (3000\$)	April 2025
UofT Dean's List Scholar Designation	June 2023 – Present
Later Life Learning OSOTF Award (500\$) (x2)	June 2023 and June 2025
University of Toronto Engineering Science Entrance Scholarship	March 2022
Bronze Governor General's Award	June 2022

PRIOR POSITIONS

Senior Research Assistant <i>Social AI Lab (PI: William Cunningham)</i> * Leading an experiment focused on determining people's preferences when solving problems: follow their surrounding's behaviour or derive the optimal solution themselves * Help with the development of Sorrel, a multi-agent reinforcement learning framework * Strongly improve the laboratory's vibrant culture by leading social and academic activities (paper discussions)	Sept. 2025 – Present Toronto, Ontario
Senior Research Assistant <i>Toronto Experimental Economics Lab (PI: Yoram Halevy)</i> * Help efficiently run in-person experiments * Designed an experiment to determine whether Kahneman et al.'s (1991) endowment effect ultimately stems from uncertainty aversion	March 2025 – Present Toronto, Ontario
Senior Research Assistant <i>Centre for Mind and Morality (PI: Paul Bloom)</i> * Schedule and test young children and adults in-person and online using Qualtrics, Prolific, and Salesforce (surveys and looking-time paradigm) * Develop a new baby-looking time software using Python * Analyze and refine participant data collection and coding procedures * Set up and worked with the EyeLink 1000 Plus * Strongly improve the laboratory's vibrant culture by leading social activities	Sept. 2023 – Sept. 2026 Toronto, Ontario
Research Practicum Student <i>Einstein Lab (PI: Gillian Einstein)</i> * Help with processing diffusion tensor imaging data to determine the relationship between estradiol intake and the organization of the brain's white matter microstructures	January 2026 – April 2026 Toronto, Ontario
Inbound and Outbound Volunteer <i>Distress Centres of Greater Toronto</i> * Answered calls from people in extreme immediate distress * Deliver personalized outbound calls to citizens who are in need of emotional support, safety check-ins, and/or medication reminders * Provide active and empathetic listening support without judgment * Extensively trained in client-centered therapy with solution-focused approach that addresses suicide, abuse, violence, self-harm, crisis support, and risk assessment	Sept. 2023 – Sept. 2025 Toronto, Ontario

Economics Thesis 1: Self-defense Legislature Preferences and Demographic Characteristics

- * Examined how individual differences in self-construal, perceived safety, and institutional trust predict preferences for these two legislative approaches across two crime contexts
- * Designed a randomized within-subjects experiment measuring self-construal, safety, trust in governing entities, and preferences over communities varying in self-defense laws and violent crime rates
- * Created an exploded logit model with maximum likelihood estimation to process strict preference ordering data; Proposed a mixed logit model with MLE
- * Awarded Economics Undergraduate Research Grant to support this project
- * Repository: github.com/Andrew01221/ECO499

Economics Thesis 2: Causal Structure Learning

- * TBA

Psychology Thesis: Social Simulation

- * TBA

Causal RL with Structural Learning

- * TBA

Bayesian Model for Selective Learning in Children

- * Designed and conducted an experiment with adults who pretended to be 12-year-old children (N=11) to investigate selective social learning using a novel ball-and-urn paradigm, built as a custom Streamlit application
- * Developed a Bayesian decision-theoretic model of selective social learning, incorporating Beta-Binomial belief updating, mixture distributions, and expected utility maximization over personal and social sampling actions
- * Fit model parameters to empirical data, recovering qualitative patterns in children's sampling behaviour across accurate and inaccurate social agent conditions
- * Repository: github.com/Andrew01221/PSY319

A Bayesian Replication of the Trade-off Between Cognitive Cost and Social Agent Accuracy

- * Developed a Bayesian computational model of social learning, formalizing the trade-off between personal cognitive cost and peer accuracy in a ball-and-urn estimation paradigm
- * Implemented numerical simulations in Python modeling an agent's sequential sampling decisions under uncertainty, integrating Beta-Binomial belief updates with expected utility maximization
- * Demonstrated model validity across same-urn and different-urn conditions, recovering qualitative predictions consistent with empirical findings on selective social learning in humans
- * Repository: github.com/Andrew01221/APM348

Moral Judgments of Schadenfreude

- * Developed a structural equation model capturing schadenfreude intensity and its moral evaluation by translating related psychological constructs into formal mathematical representations
- * Designed both a naturalistic experiment and a factorial design to validate the model
- * Awarded Undergraduate Research Fund to support this project
- * Repository: github.com/Andrew01221/PSY405

Demyelination and Disorganization of White Matter Tissue in Aging Populations

- * Analyzed diffusion tensor imaging (DTI) data from transfeminine adults, cisgender women, and cisgender men (N=28) using FSL's tract-based spatial statistics (TBSS) pipeline to assess white matter microstructure organization
- * Conducted voxelwise one-way ANOVAs with permutation-based correction for multiple comparisons across the white matter skeleton to examine group differences in fractional anisotropy
- * Quality checked structural MRI data (T1, T2, and DTI)

* Repository: github.com/Andrew01221/PSY309

Determining the Provenance of Higher Willingness to Pay in the American Housing Market

- * Wanted to determine whether status quo bias or the endowment effect contribute to a higher willingness to accept in the American housing market
- * Constructed a 2015-2023 panel dataset from the American Housing Survey (around 500,000 observations)
- * Operationalized status quo bias using the homeowner's number of self-reported daily difficulties
- * Operationalized the endowment effect using homeowner tenure
- * Used a two-way fixed effects model with my status quo bias and endowment effect variables
- * Found that only status quo bias is associated with a higher willingness to accept
- * Repository: github.com/Andrew01221/ECO475

Investigating Erroneous Theory of Mind in the Context of Speed-Dating

- * Found that overconfidence bias and social projection bias heavily mediate a person's accuracy when determining if a partner is romantically interested in them
- * Data used: 4913 match-level speed-dating observations
- * Used logistic regression with ridge and lasso regularizations, KNN, Random Forests, and XGBoost with hyperparameter tuning and 5-fold cross-validation for statistical analysis
- * Github link: <https://github.com/Andrew01221/ECO482>

Exploring the Impact of Economic Factors on Sentiment: A Test of Economic Voting Theory

- * Found a significant relationship between how often people posted negative tweets about Trump and county-level economic factors during the 2020 presidential election
- * Data used: 1.8 million 2020 political tweets targeted at Trump and Biden and period-relevant county and state-level economic conditions
- * Used pandas, geopandas, BeautifulSoup, RandomForestRegressor, matplotlib, and TextBlob for data cleaning, geographical mapping, web scraping, statistical modeling, and sentiment analysis, respectively
- * Github link: <https://github.com/Andrew01221/ECO225>

Social Proof Theory as a Decision-making Mechanism in Online Negotiations

- * Found that social proof theory is present when negotiating item prices on eBay
- * Developed a multiple linear regression model to examine the effect of external social influences—the price of a previously sold similar item and the seller's starting price—on a consumer's behavioural output—the buyer's initial offering price, using STATA
- * GitHub Link: <https://github.com/Andrew01221/ECO375>

Preferential Distribution of Harmful and Beneficial Experiences across a Population

- * Determined how people prefer to distribute harmful and beneficial experiences across a neutral and homogeneous population
- * Created a population-level stochastic choice model that uses strict preference orderings with proof
- * GitHub Link: <https://github.com/Andrew01221/PSY299>

CONFERENCE PRESENTATIONS

Scutt, A. (2024). Preferential Distributions of Harmful and Beneficial Experiences Across a Population. Poster presented at the Research Opportunities Program Poster Fair at the University of Toronto, Toronto, ON.

Scutt, A. (2024). Preferential Distributions of Harmful and Beneficial Experiences Across a Population. Flash talk presented at the Psychology Undergraduate Research Club at the University of Toronto, Toronto, ON.

RELEVANT COURSEWORK

University of Washington: Computational Neuroscience

DeepLearning AI: Unsupervised Learning, Recommenders, Reinforcement Learning

DeepLearning AI: Deep Learning Specialization (5 Courses) coupled with Deep Learning textbook by Ian Goodfellow, Yoshua Bengio, and Aaron Courville

Stanford University: Introduction to Logic

MAT337: Introduction to Real Analysis

STA447: Stochastic Processes

APM462: Nonlinear Optimization

RELEVANT SKILLS

Languages and Framework: Qualtrics, Prolific, Salesforce, Python (pandas, NumPy, scikit, geopandas, BeautifulSoup, TextBlob, sklearn, TextBlob, Sorrel, PsychoPy, and tensorflow), R, STATA, Django, Java, JavaScript, Bash, PowerShell, SQL, React, Redux, HTML, CSS, JASP, MatLab, and Kali Linux

Personal Skills: Perseverant, Perceptive, Creative, Curious, and Positive attitude

Github: <https://github.com/Andrew01221>

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